

File: cm5_socket.kicad_sch

File: system_mcu.kicad_sch

File: ac_dc_power.kicad_sch

File: digital_io_fan_encoder.kicad_sch

File: usb3_hub.kicad_sch

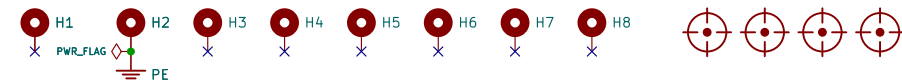
File: usb_conn.kicad_sch

File: video_outputs.kicad_sch

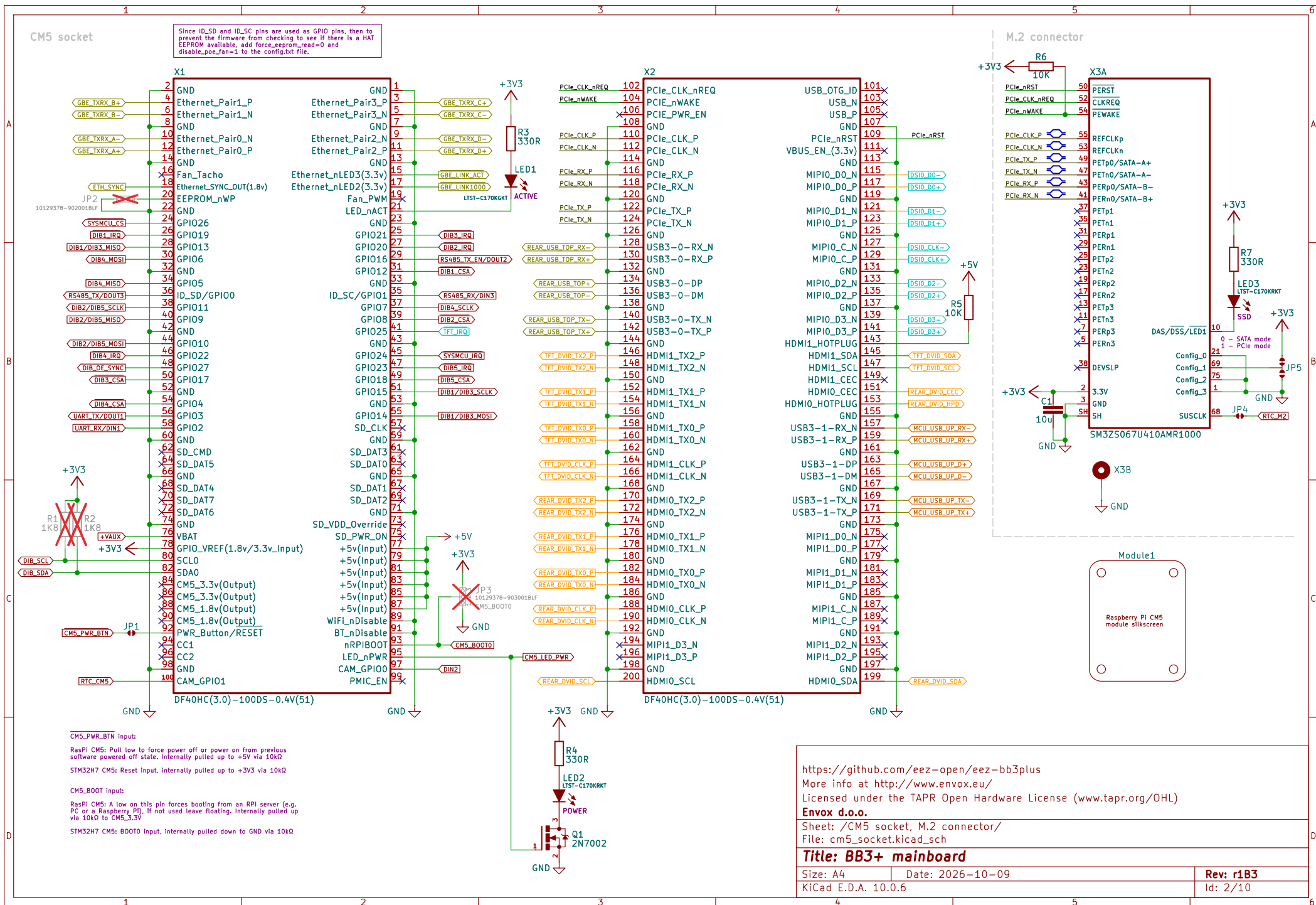
File: dib.kicad_sch

File: power_coupling.kicad_sch

I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x7A	PTN5150A USB CC controller



Size: A4	Date: 2026-10-09	Rev: r1B3
KiCad E.D.A. 10.0.6		Id: 1/10



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /CM5 socket, M.2 connector/

File: cm5_socket.kicad_sch

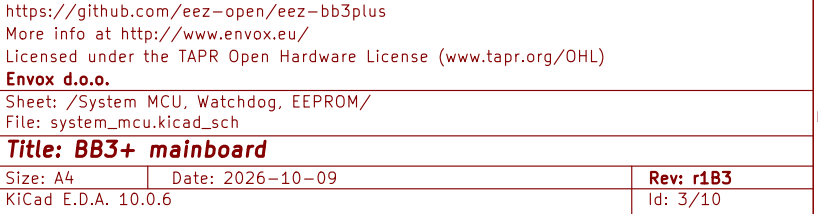
Title: BB3+ mainboard

Size: A4 Date: 2026-10-09

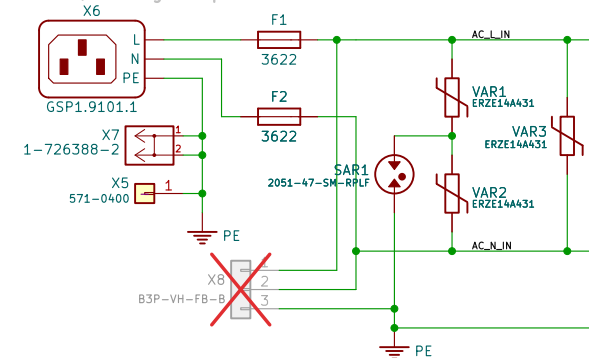
KiCad E.D.A. 10.0.6

Rev: r1B3

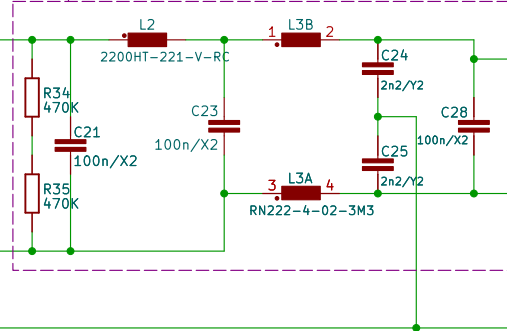
Id: 2/10



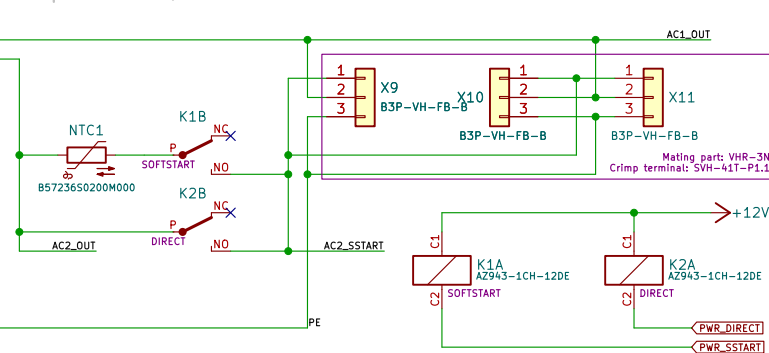
AC inlet, filtering and protection



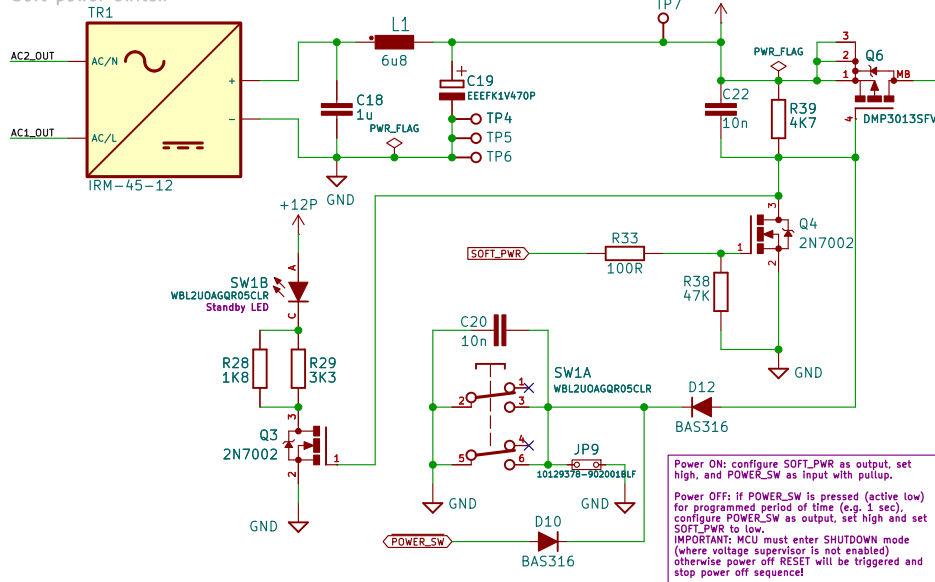
EMI compliance filter



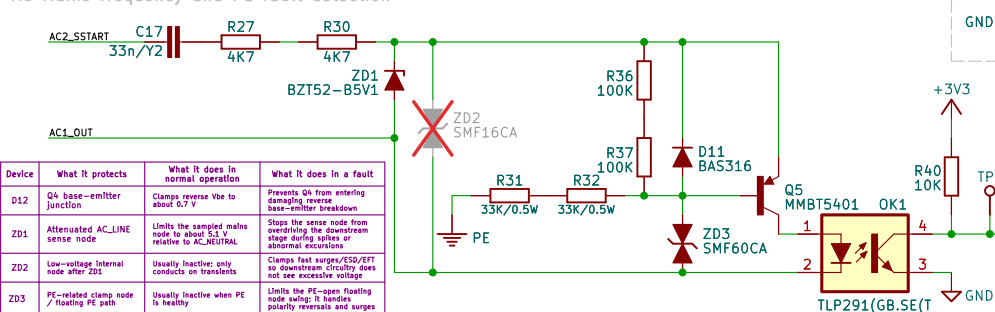
AC power control, Inrush current limiter



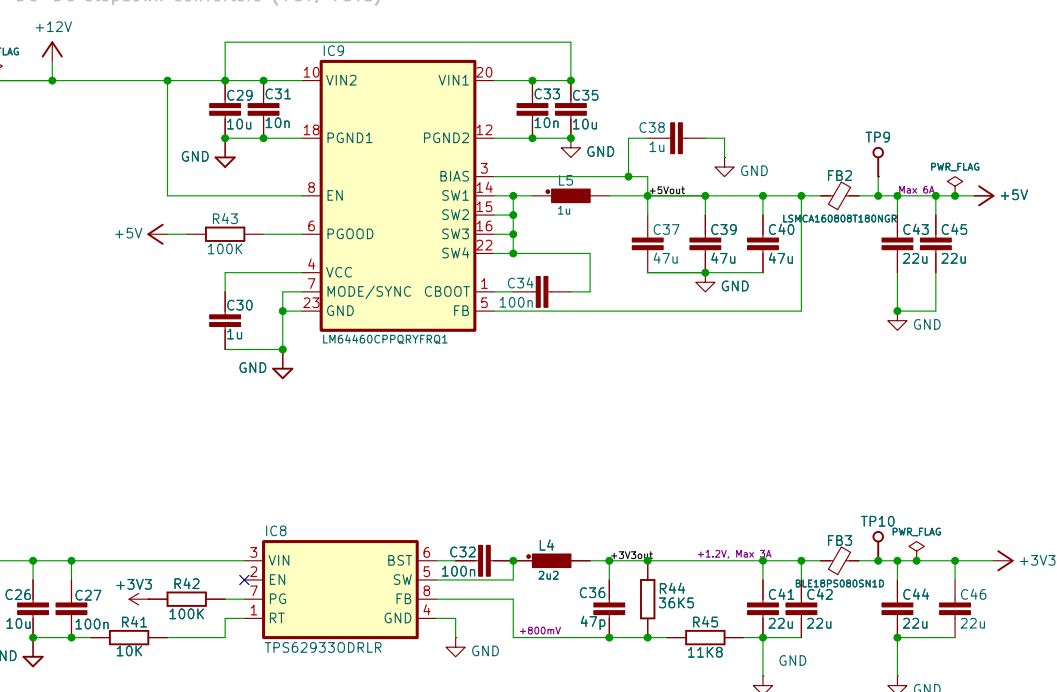
Soft power switch



AC Mains frequency and PE fault detection



DC-DC stepdown converters (+5V, +3V3)



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Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

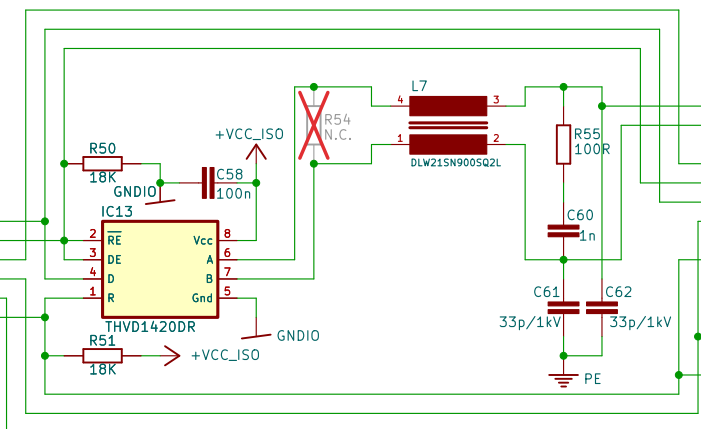
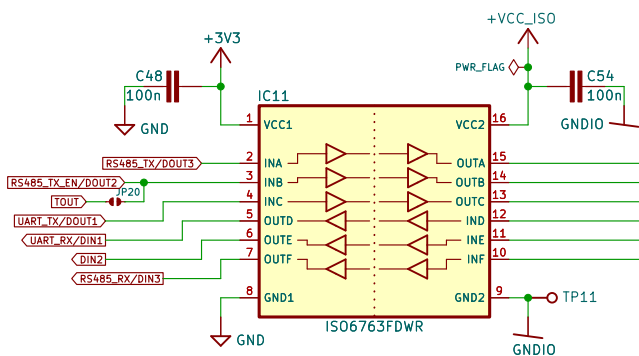
Size: A4 Date: 2026-10-09

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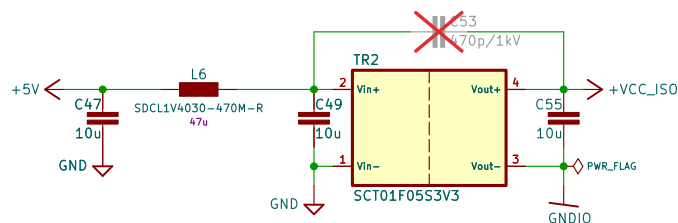
Isolated digital I/O & RS485 transceiver (half-duplex)



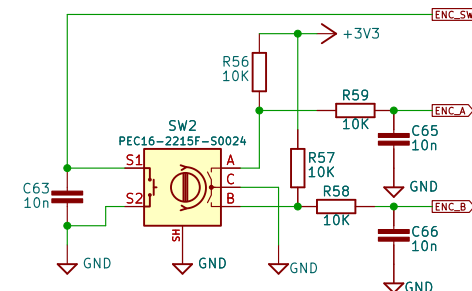
Rear panel I/O connector

- +VCC_ISO 1
- RS485_A_ISO 2
- RS485_B_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

Isolated regulated power supply (+5V)



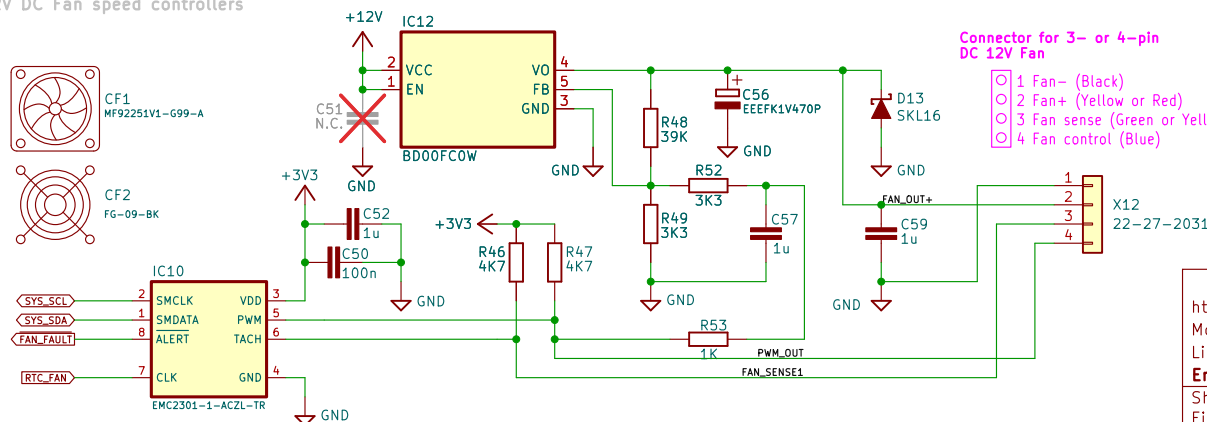
Encoder with switch



Encoder shaft length: 15 mm / Flatted

Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)

12V DC Fan speed controllers



Connector for 3- or 4-pin DC 12V Fan

- 1 Fan- (Black)
- 2 Fan+ (Yellow or Red)
- 3 Fan sense (Green or Yellow)
- 4 Fan control (Blue)

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

Title: BB3+ mainboard

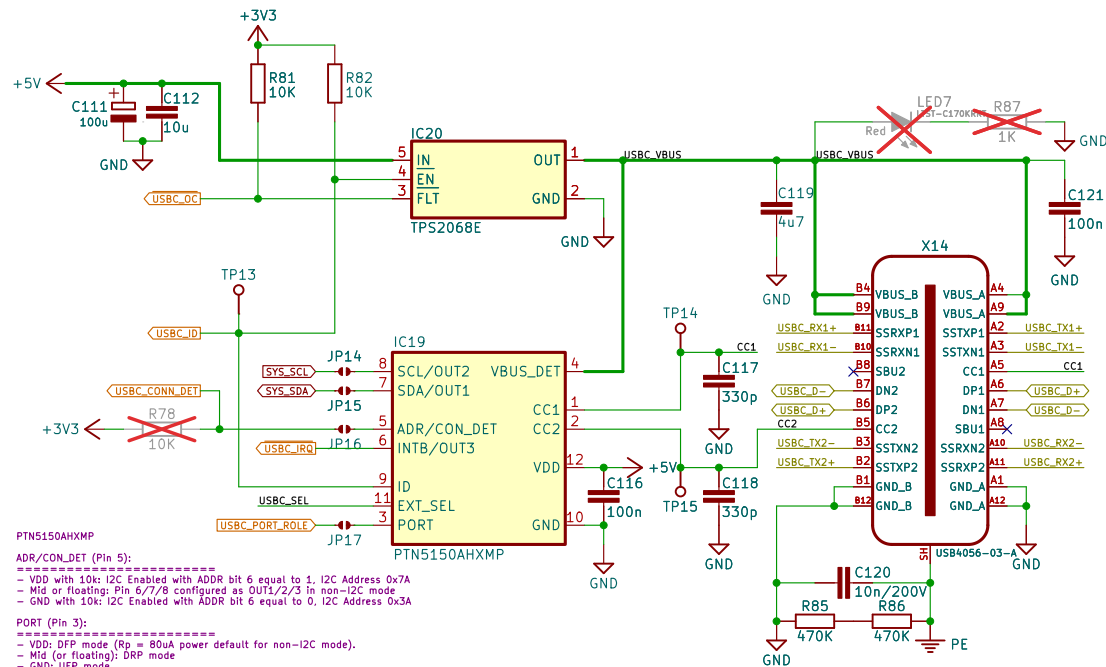
Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

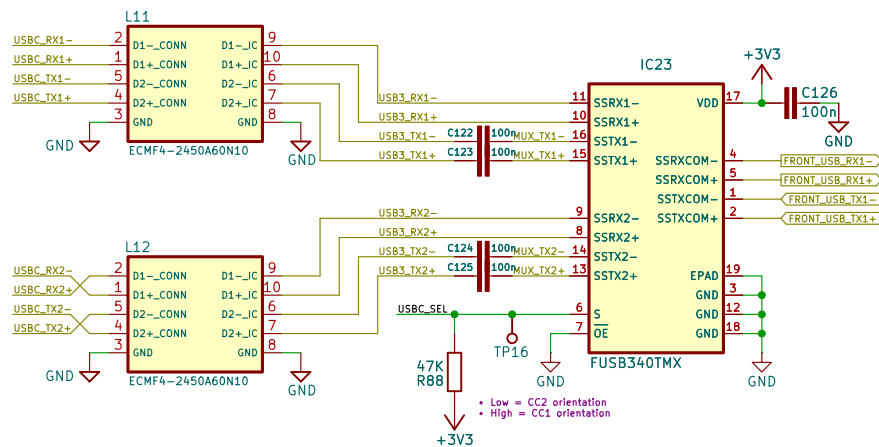
Rev: r1B3

Id: 5/10

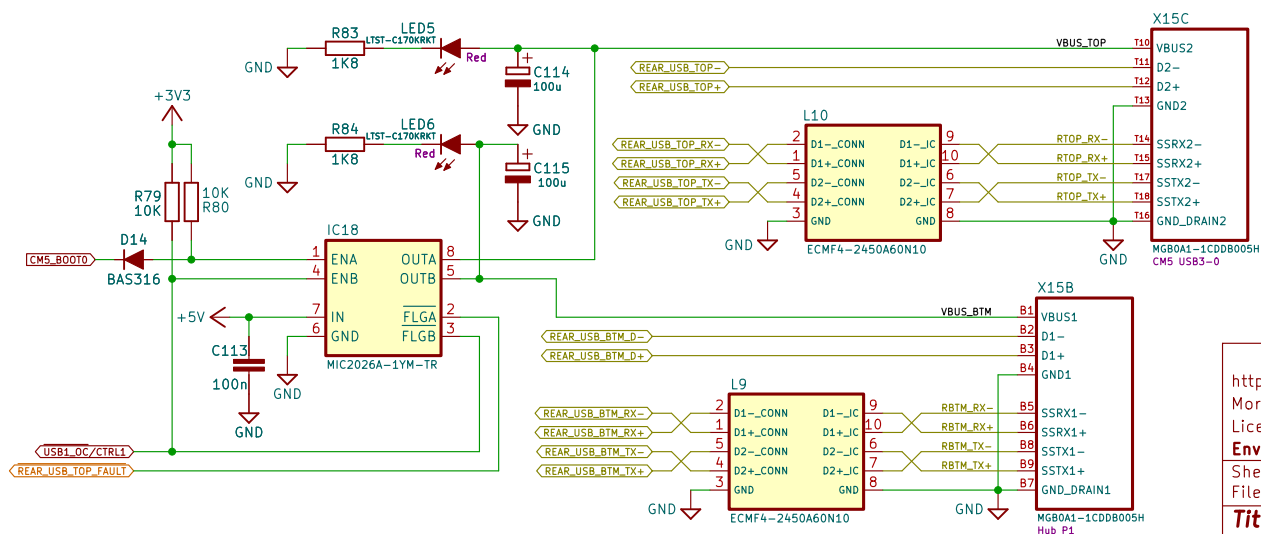
USB-C 3.0 front port (source max 1.5A)



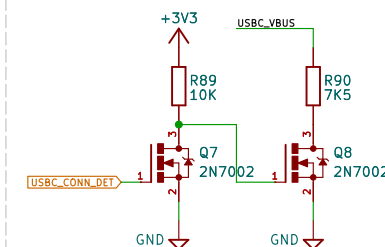
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



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Envox d.o.o.

Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

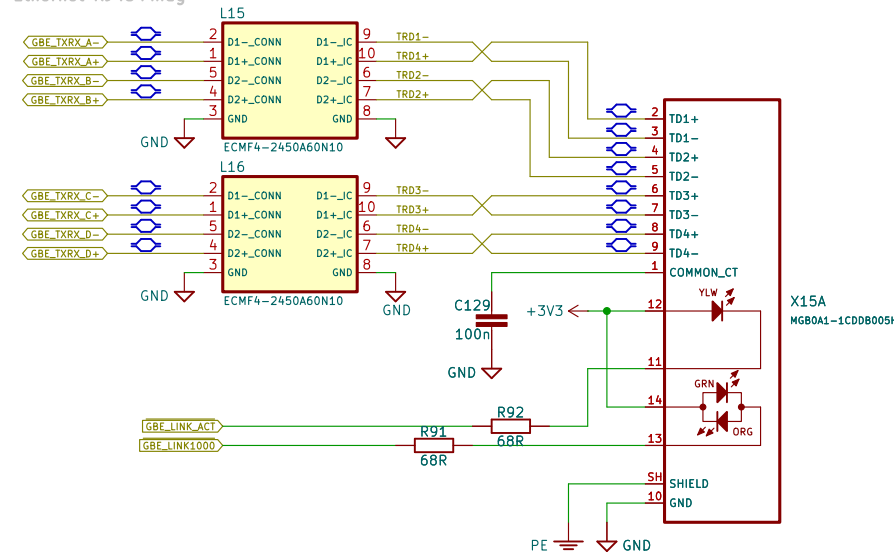
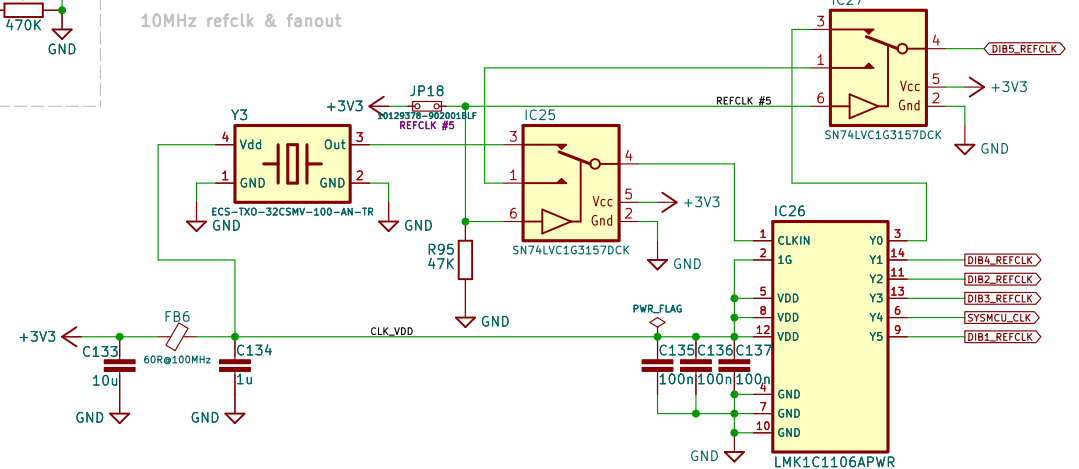
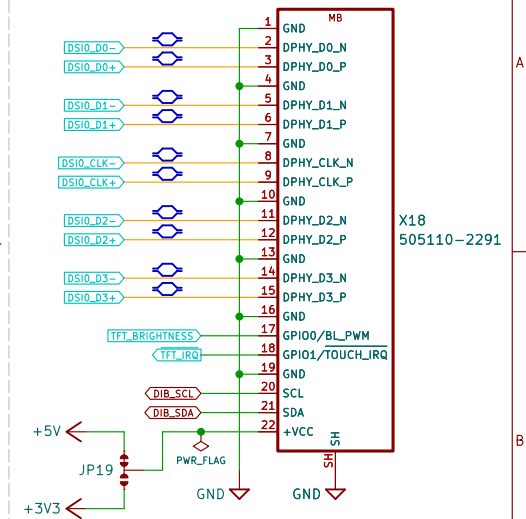
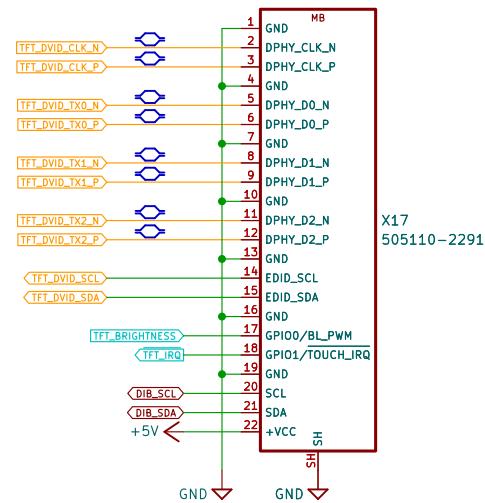
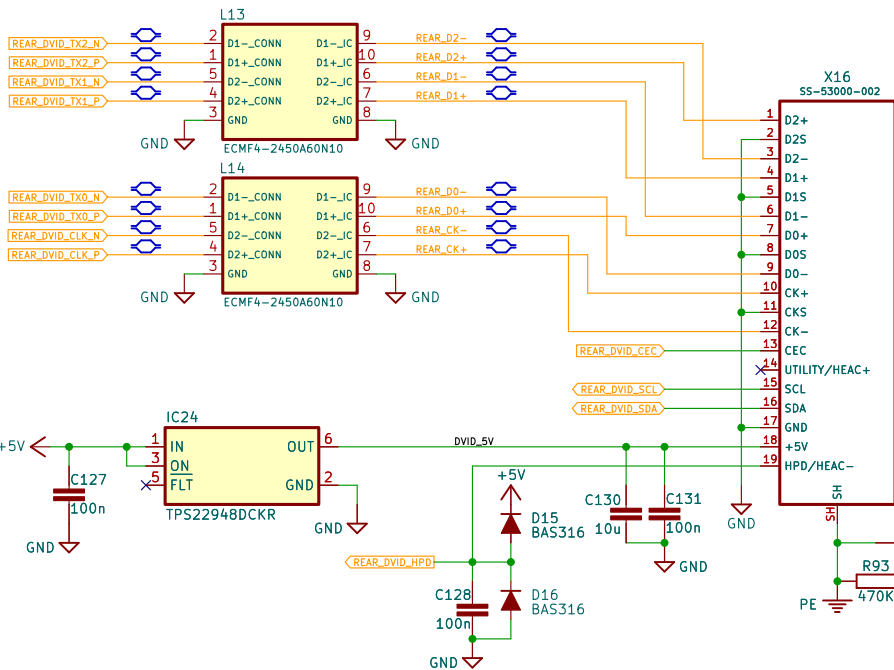
Title: BB3+ mainboard

Size: A4 Date: 2026-10-09

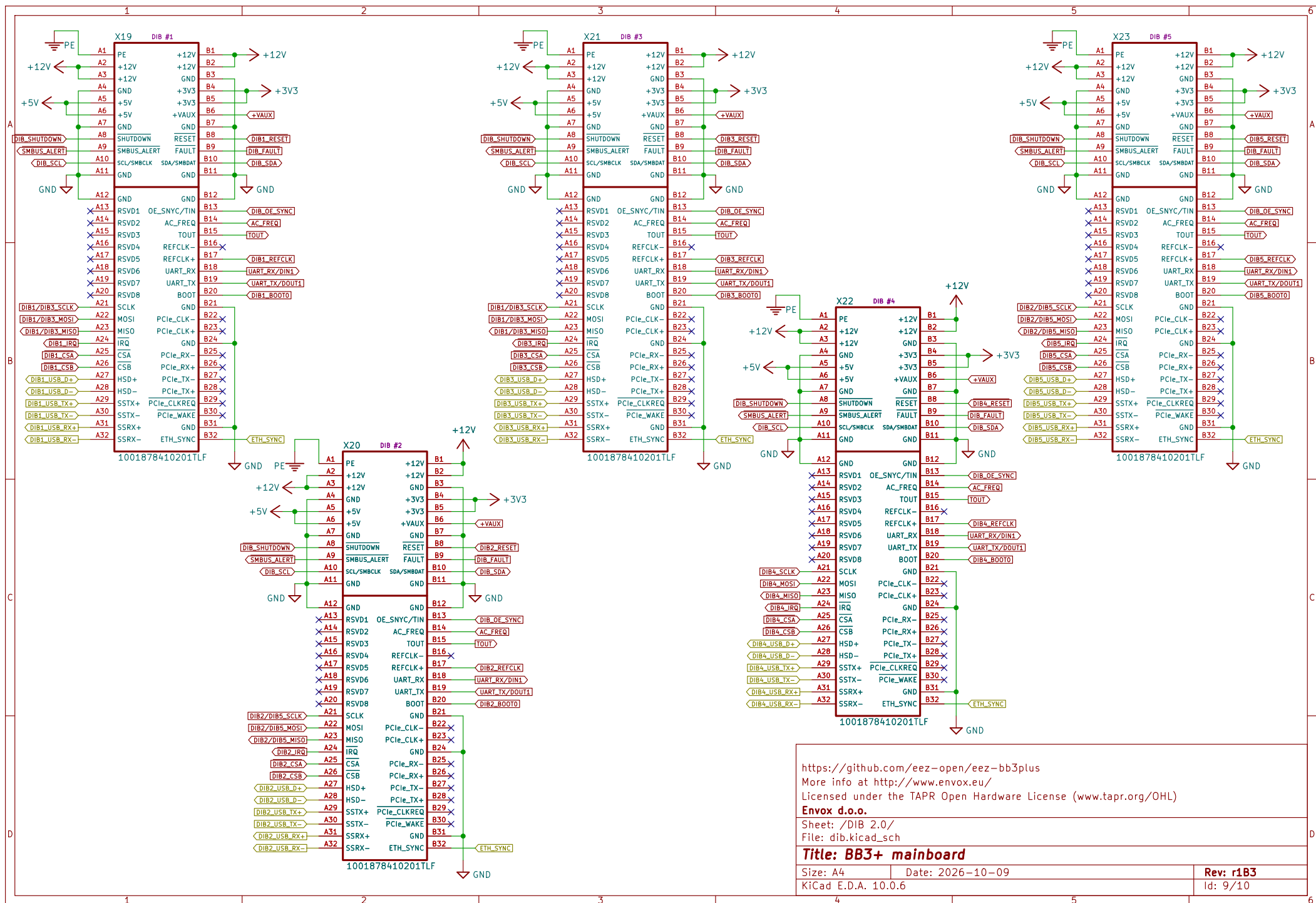
KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 7/10



Id: 8/10



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Envox d.o.o.

Sheet: /DIB 2.0/

File: dib.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

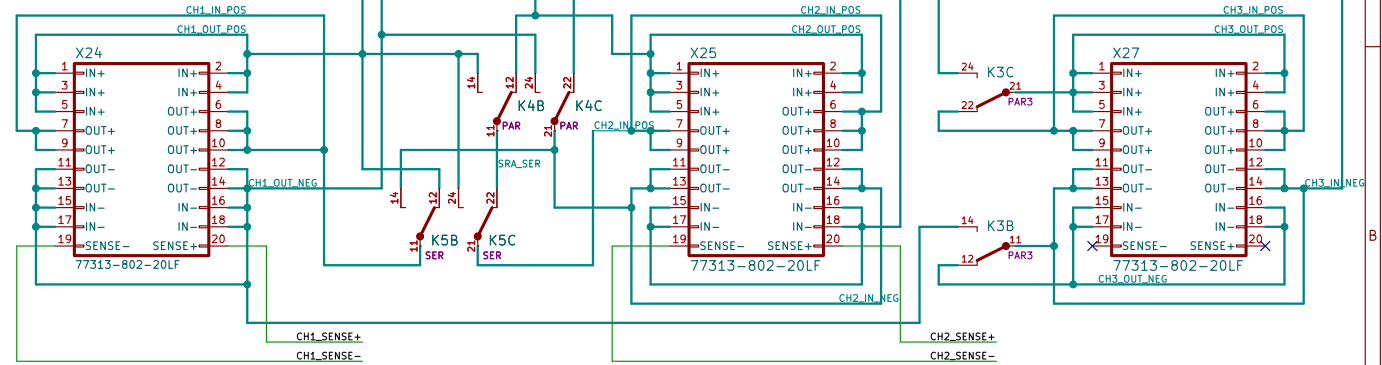
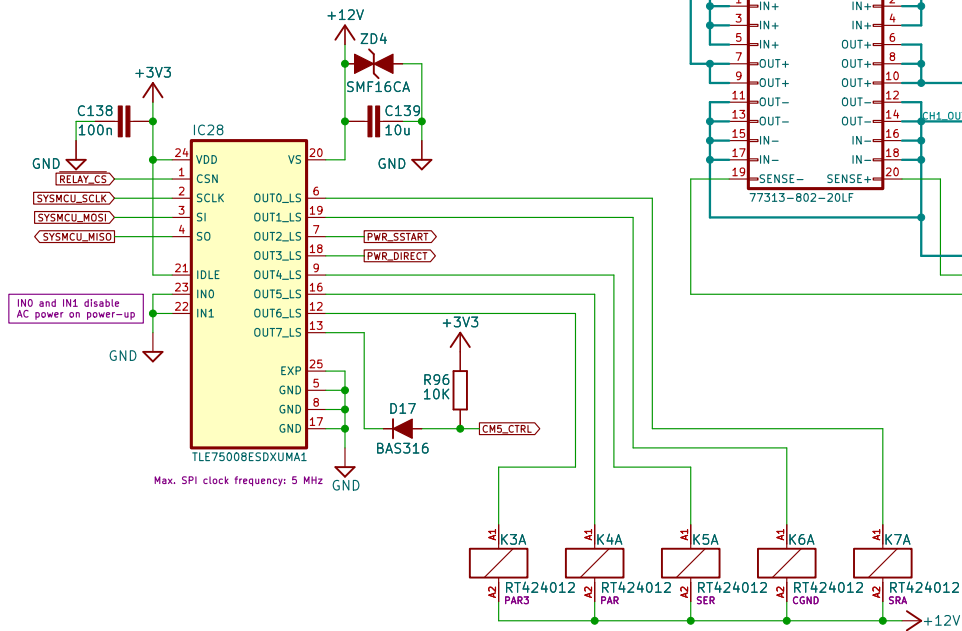
Rev: r1B3

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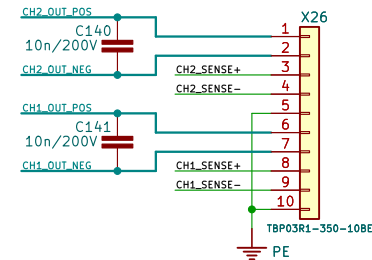
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 2	3
CH2_SENSE+ 3	4
CH2_SENSE- 4	5
PE 5	6
CH1_OUT+ 6	7
CH1_OUT- 7	8
CH1_SENSE+ 8	9
CH1_SENSE- 9	10
PE 10	

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10